



Department of Information and Communication Technology

Faculty of Technology

University of Ruhuna

Object Oriented Programming Practicum

ICT 2132

Mini Project 2026 B09

Group 07

Submitted to: Mr.P.H.P. Nuwan Laksiri

Submitted by: TG/2024/2077 Vishwa Pramuditha

Group Members

Group – 07

Members –

No	Index Number	Name
1	TG/2024/2069	T.T.Ranathunga
2	TG/2024/2070	W.A.P.Methmadu
3	TG/2024/2077	R.A.V.Pramuditha
4	TG/2024/2078	B.A.N.Nethmina

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1. Introduction

1.1 Purpose

This Software Requirements Specification defines the functional and non-functional requirements of the proposed Faculty of Technology Academic Management System. The system provides a centralized solution for managing user profiles, courses, course materials, marks, grades, GPA, attendance, medical records, notices, timetables and examination eligibility.

1.2 Document Conventions

SHALL / MUST – mandatory requirement. SHOULD – recommended requirement. MAY – optional requirement. FR – Functional Requirement. NFR – Non-Functional Requirement.

1.3 Product Scope

The system provides role-based academic information management. Admins manage users, courses, notices and timetables. Lecturers manage course materials and marks and access relevant undergraduate information. Technical Officers manage attendance and medical records.

Undergraduates access their own courses, attendance, medical records, grades, GPA, timetable and notices.

1.5 References

ICT2132 – Object Oriented Programming Practicum Mini Project – B09 specification; UML use-case/class concepts; MySQL database concepts; Java OOP concepts; and the UGC Commission Circular No. 12-2024 requirement identified in the project brief.

2. Overall Description

2.1 Product Perspective

The Faculty of Technology Academic Management System is a centralized academic information management application intended to reduce dependence on manual academic record management.

2.2 Main Actors

Actor	Main Responsibilities
Admin	Manage users, courses, notices and timetables.
Lecturer	Manage profile, course materials, marks and access undergraduate academic information.
Technical Officer	Manage attendance and medical records; view notices and department timetables.
Undergraduate	Update contact/profile information and view courses, attendance, medicals, grades, GPA, timetable and notices.

2.3 Product Functions

User Management: authentication, user creation, updating, profiles and roles.

Course Management: course creation, modification, viewing, assignment and materials.

Attendance: recording, updating, percentage calculation and theory/practical/combined views.

Medical: submissions, approval status and maintenance.

Marks and Grades: mark entry, eligibility, final marks, grades, grade points, SGPA and CGPA.

Notices and Timetable: creation, modification, publishing and viewing.

2.4 User Characteristics

Admins are expected to understand faculty administration and basic computer operations. Lecturers should be able to enter academic marks and access undergraduate academic information. Technical Officers should be able to manage attendance and medical information. Undergraduates should have basic computer literacy.

2.5 Operating Environment

Software: Java/JDK, MySQL Server, JDBC/MySQL Connector, Java Swing or JavaFX, Windows/Linux.

Hardware: dual-core processor or better, 4 GB RAM or better, 10 GB available storage, display, keyboard and mouse.

3. System Scope

3.1 In-Scope

Authentication; user profiles; course management; course materials; marks; grades; GPA; attendance; medical records; examination eligibility; notices; and timetables.

3.2 Out-of-Scope

Online payments, hostel management, library management, payroll, salary management, online learning/video conferencing, admission processing and automated examination paper generation.

4. Stakeholders and User Classes

Stakeholder	Interest
Faculty Administration	Accurate and controlled academic information.
Admin	System and master-data management.
Lecturer	Course and undergraduate academic management.
Technical Officer	Attendance and medical management.
Undergraduate	Access to personal academic information.

5. System Features

5.1 Authentication

Secure login using username/password, role identification and logout.

5.2 User Profile Management

Maintain profiles for Admin, Lecturer, Technical Officer and Undergraduate.

5.3 Course Management

Create, update, view and delete course information.

5.4 Course Material Management

Lecturers add and modify notes, practical materials, documents and references.

5.5 Attendance Management

Record theory and practical attendance, update records and calculate percentages.

5.6 Medical Records

Store medical submissions, dates, course/session information and approval status.

5.7 Marks Management

Upload and validate marks, maintain CA/final marks and determine eligibility.

5.8 Grade Calculation

Calculate grades and grade points using the defined grading scheme.

5.9 SGPA and CGPA

Calculate semester and cumulative grade point averages using credits and grade points.

5.10 Notice Management

Admin creates, edits, deletes and publishes notices.

5.11 Timetable Management

Admin creates and maintains timetable entries.

6. Functional Requirements

6.1 Authentication Requirements

FR-001 – The system shall allow registered users to log in.

FR-002 – The system shall validate username and password.

FR-003 – The system shall deny access when invalid credentials are entered.

FR-004 – The system shall identify the user's role after authentication.

FR-005 – The system shall provide role-based access control.

FR-006 – The system shall allow users to log out.

6.2 Admin Requirements

FR-007 – Admin shall create user profiles.

FR-008 – Admin shall update user profiles.

FR-009 – Admin shall delete user profiles.

FR-010 – Admin shall view user profiles.

FR-011 – Admin shall create courses.

FR-012 – Admin shall update courses.

FR-013 – Admin shall delete courses.

FR-014 – Admin shall view courses.

FR-015 – Admin shall create notices.

FR-016 – Admin shall update notices.

FR-017 – Admin shall delete notices.

FR-018 – Admin shall create timetable entries.

FR-019 – Admin shall update timetable entries.

FR-020 – Admin shall delete timetable entries.

6.3 Lecturer Requirements

FR-021 – Lecturer shall log in.

FR-022 – Lecturer shall update their profile.

FR-023 – Lecturer shall not change username through normal profile editing.

FR-024 – Lecturer shall not change password through normal profile editing.

FR-025 – Lecturer shall add course materials.

FR-026 – Lecturer shall modify course materials.

FR-027 – Lecturer shall upload examination marks.

FR-028 – Lecturer shall view undergraduate details.

FR-029 – Lecturer shall view undergraduate eligibility.

FR-030 – Lecturer shall view undergraduate marks.

FR-031 – Lecturer shall view grades.

FR-032 – Lecturer shall view GPA.

FR-033 – Lecturer shall view attendance.

FR-034 – Lecturer shall view medical records.

FR-035 – Lecturer shall view notices.

6.4 Technical Officer Requirements

FR-036 – Technical Officer shall log in.

FR-037 – Technical Officer shall update their profile.

FR-038 – Technical Officer shall not modify username through normal profile editing.

FR-039 – Technical Officer shall not modify password through normal profile editing.

FR-040 – Technical Officer shall add attendance records.

FR-041 – Technical Officer shall update attendance records.

FR-042 – Technical Officer shall maintain undergraduate attendance.

FR-043 – Technical Officer shall add medical records.

FR-044 – Technical Officer shall update medical records.

FR-045 – Technical Officer shall maintain medical information.

FR-046 – Technical Officer shall view notices.

FR-047 – Technical Officer shall view department timetables.

6.5 Undergraduate Requirements

FR-048 – Undergraduate shall log in.

FR-049 – Undergraduate shall update contact details.

FR-050 – Undergraduate shall update profile picture.

FR-051 – Undergraduate shall not modify restricted profile information.

FR-052 – Undergraduate shall view attendance.

FR-053 – Undergraduate shall view medical details.

FR-054 – Undergraduate shall view courses.

FR-055 – Undergraduate shall view grades.

FR-056 – Undergraduate shall view GPA.

FR-057 – Undergraduate shall view timetable.

FR-058 – Undergraduate shall view notices.

6.6 Attendance Requirements

FR-059 – Store attendance for each undergraduate and course.

FR-060 – Support 15 theory sessions for the required dataset.

FR-061 – Support 15 practical sessions for the required dataset.

FR-062 – Calculate attendance percentages.

FR-063 – Display individual attendance.

FR-064 – Display batch attendance summaries.

FR-065 – Support theory-only checking.

FR-066 – Support practical-only checking.

FR-067 – Support combined checking.

FR-068 – Identify attendance above 80%.

FR-069 – Identify attendance equal to 80%.

FR-070 – Identify attendance below 80%.

FR-071 – Associate medical records with relevant attendance dates.

6.7 Medical Requirements

FR-072 – Store medical records.

FR-073 – Identify the undergraduate associated with a medical.

FR-074 – Record medical date.

FR-075 – Record approval status.

FR-076 – Allow authorized users to update medical records.

FR-077 – Distinguish approved and unapproved medical records.

6.8 Marks Requirements

FR-078 - Store marks for each course.

FR-079 – Store marks out of 100.

FR-080 – Lecturer shall upload marks.

FR-081 – Validate marks before saving.

FR-082 – Reject marks outside the valid range.

FR-083 – Calculate CA eligibility.

FR-084 – Identify final-examination eligibility.

FR-085 – Display marks individually.

FR-086 – Display batch marks.

6.9 Grade Requirements

FR-087 – Calculate grades using the defined grading scheme.

FR-088 – Display grades for each course.

FR-089 – Calculate grade points.

FR-090 – Calculate SGPA.

FR-091 – Calculate CGPA.

FR-092 – Display individual results.

FR-093 – Display batch results.

7. Non-Functional Requirements

ID	Requirement
NFR-001	The system should respond to normal actions within an acceptable period.
NFR-002	Database queries should be optimized to avoid unnecessary delays.
NFR-003	The system should efficiently handle the required number of users and records.
NFR-004	Protected functionality shall require authentication.
NFR-005	Passwords shall not be stored in plain text.
NFR-006	Role-based authorization shall be implemented.
NFR-007	Users shall only access functions permitted for their roles.
NFR-008	Unauthorized users shall not access restricted academic information.
NFR-009	Database access shall use appropriate authentication.
NFR-010	The user interface shall be easy to understand.
NFR-011	Menus shall be organized according to user roles.
NFR-012	Error messages shall be understandable.
NFR-013	The system shall maintain data consistency and prevent invalid data.
NFR-014	The software shall follow modular object-oriented design.
NFR-015	The application should run on supported Java/MySQL environments.
NFR-016	The database shall allow growth without major redesign.

8. External Interface Requirements

8.1 User Interface

The application shall provide graphical user interfaces appropriate to each role.

- Login
- Admin: User Management | Course Management | Notice Management | Timetable | Logout
- Lecturer: Profile | Courses | Materials | Marks | Undergraduate Details | Eligibility |
- Attendance | Medicals | Grades/GPA | Notices | Logout
- Technical Officer: Profile | Attendance | Medicals | Department Timetable | Notices | Logout
- Undergraduate: Profile | Courses | Attendance | Medicals | Grades | GPA | Timetable | Notices | Logout

8.2 Hardware Interface

The system shall use standard computer hardware including keyboard, mouse, display and storage. No specialized hardware is required.

8.3 Software Interface

Java: application development.

MySQL: persistent storage.

JDBC: communication between Java and MySQL.

9. Use Case Model and Specifications

9.1 Major Use Cases

- Admin -> Manage Users, Manage Courses, Manage Notices, Manage Timetable
- Lecturer -> Update Profile, Manage Materials, Upload Marks, View Undergraduates, Check Eligibility, View Marks/Grades/GPA, View Attendance, View Medicals
- Technical Officer -> Manage Attendance, Manage Medicals, View Notices, View Timetable
- Undergraduate -> Update Contact/Profile Picture, View Attendance, Medicals, Courses, Grades/GPA, Timetable and Notices

UC-01: Login s

Actors: Admin, Lecturer, Technical Officer, Undergraduate.

Preconditions: registered account and available system.

Flow: open login, enter credentials, validate, identify role and display dashboard. Invalid credentials produce an error and allow retry.

UC-02: Manage User Profiles

Actor: Admin.

Flow: open User Management, view users, create/edit/delete, validate, save and confirm.

Errors include duplicate username and missing required fields.

UC-03: Upload Marks

Actor: Lecturer.

Preconditions: course access.

Flow: select course, view undergraduates, enter marks, validate, save and confirm.

Values outside 0–100 are rejected.

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UC-04: Manage Attendance

Actor: Technical Officer.

Flow: select course/date/session, display undergraduates, record attendance, save and recalculate percentage.

UC-05: Manage Medical Record

Actor: Technical Officer.

Flow: select undergraduate, enter medical information, select approval status, validate and save.

UC-06: Check Eligibility

Actor: Lecturer.

Flow: select course/batch, retrieve attendance and CA marks, apply rules and display eligibility.

UC-07: View Academic Results

Actor: Undergraduate.

Flow: open results, retrieve marks/grades/GPA and display performance.

UC-08: Manage Courses

Actor: Admin.

Flow: add/edit/delete course information, validate and save.

UC-09: Manage Notices

Actor: Admin.

Flow: create/edit/delete/publish notice; authorized users can view it.

UC-10: Manage Timetable

Actor: Admin.

Flow: create/update/delete timetable entries and make relevant entries viewable.

10. Business Rules

Rule	Description
BR-001	Each user shall have a defined role: Admin, Lecturer, Technical Officer or Undergraduate.
BR-002	Undergraduates may update contact details and profile picture among the specified editable fields.
BR-003	Lecturers may update profiles but not username/password through normal profile editing.
BR-004	Technical Officers may update profiles but not username/password through normal profile editing.
BR-005	The system shall recognize the 80% attendance threshold.
BR-006	CA $\geq$ 40% means eligible for the final examination; CA $<$ 40% means not eligible.
BR-007	Marks shall be within 0–100.
BR-008	The required dataset shall contain 15 theory sessions per subject.
BR-009	The required dataset shall contain 15 practical sessions per subject.
BR-010	Medical records shall have an approval state.
BR-011	GPA shall use course credits and grade points.
BR-012	Grades shall follow the grading scheme specified for the project.

11. Data Requirements

Entity	Representative Attributes
User	UserID, Username, Password, Name, Email, ContactNumber, Role, Department, ProfilePicture
Undergraduate	UndergraduateID, UserID, RegistrationNumber, Batch, Department, ContactDetails, ProfilePicture
Lecturer	LecturerID, UserID, Name, Department, Email, ContactNumber
Technical Officer	TechnicalOfficerID, UserID, Name, Department, Email, ContactNumber
Course	CourseID, CourseCode, CourseName, Credits, TheoryCredits, PracticalCredits, Department, Semester
Course Material	MaterialID, CourseID, LecturerID, Title, Description, FilePath, UploadDate
Attendance	AttendanceID, UndergraduateID, CourseID, Date, SessionType, Status
Medical	MedicalID, UndergraduateID, CourseID, Date, Description, DocumentPath, ApprovalStatus
Marks	MarkID, UndergraduateID, CourseID, AssessmentType, Marks
Grade	GradeID, UndergraduateID, CourseID, FinalMarks, Grade, GradePoint
Notice	NoticeID, Title, Description, Date, CreatedBy, Status
Timetable	TimetableID, CourseID, LecturerID, Department, Batch, Date, StartTime, EndTime, Venue, SessionType

12. Data Volume Requirements

Data	Minimum Requirement
Admin	1
Lecturers	5
Technical Officers	4
Undergraduates	20
Theory sessions per subject	15
Practical sessions per subject	15

13. Object-Oriented Requirements

13.1 Classes and Objects

The implementation shall use classes such as User, Admin, Lecturer, TechnicalOfficer, Undergraduate, Course, Attendance, Medical, Mark, Grade, Notice and Timetable.

13.2 Inheritance

Admin Lecturer TechnicalOfficer Undergraduate inherits from User

13.3 Encapsulation

Attributes should be private and accessed through appropriate getters and setters.

13.4 Abstraction

Common behavior may be represented through abstract classes or interfaces, while subclasses implement role-specific behavior.

13.5 Polymorphism

Common operations such as viewDashboard() can have role-specific implementations.

13.6 Exception Handling

The application shall handle invalid login, database connection failure, invalid marks, missing fields, duplicate records, invalid attendance and SQL errors.

14. Database Requirements

MySQL shall be used as the relational database management system. The database shall maintain Users, Courses, CourseMaterials, Attendance, Medical, Marks, Grades, Notices and Timetables.

The database shall enforce primary keys, foreign keys, NOT NULL constraints, unique constraints, appropriate data types and referential integrity.

15. Security Requirements

- **Authentication:** protected functions require login.
- **Authorization:** permissions are role-based.
- **Password Security:** passwords should be securely hashed rather than stored as plain text.
- **Database Security:** credentials should not be unnecessarily exposed.
- **Privacy:** academic information shall only be accessible to authorized users.

16. Validation Requirement

The system shall validate all user input. Required fields cannot be empty. Emails should follow an acceptable format. Marks must satisfy $0 \leq \text{marks} \leq 100$. Course codes and user IDs shall be unique where required. Attendance values must represent valid states.

17. Error and Exception Handling

The system shall display meaningful errors and should not terminate unexpectedly due to normal user input.

- Invalid username or password.
- Marks must be between 0 and 100.
- Required field cannot be empty.
- User already exists.
- Unable to connect to database.
- Invalid attendance record.

18. Reporting Requirements

The system shall support individual and batch-level viewing of attendance, marks, grades and GPA information. Attendance views shall include theory, practical and combined results. Results shall include marks, grades, grade points, SGPA and CGPA where applicable.

19. Acceptance Criteria

Area	Acceptance Condition
Authentication	Users can log in; invalid credentials are rejected; correct dashboards are shown.
User Management	Admin can create, update, delete and view users.
Courses	Admin can create, modify, delete and view courses.
Attendance	Technical Officer can record attendance and the system calculates percentages and required views.
Medical	Medical records can be created and approval status maintained.
Marks	Lecturer can upload marks; values are validated 0–100; CA eligibility is calculated.
Grades	Grades, grade points, SGPA and CGPA are calculated and displayed.
Notices	Admin can create notices and users can view them.
Timetable	Admin can manage timetables and relevant users can view them.

20. Required Test Data

The system shall contain sufficient sample data to demonstrate all required scenarios.

- 1 Admin
- 5 Lecturers
- 4 Technical Officers
- 20 Undergraduates

Attendance scenarios:

- More than 80%
- Exactly 80%
- Less than 80% without medical
- More than 80% with medical
- Less than 80% with medical

21. Assumptions

- Every system user has a unique username.
- Each undergraduate belongs to a batch.
- Each course belongs to an academic department.
- Lecturers are assigned to relevant courses.
- Technical Officers manage attendance and medical information.
- Admin maintains system master data.
- Academic grading follows the required grading criteria.
- The database is available when the application is used.
- Users have valid login credentials.
- Academic evaluation criteria are provided by course coordinators.

22. Dependencies

The system depends on Java/JDK, MySQL Server, JDBC driver, operating system support, database availability and correct academic evaluation criteria and timetable information.

23. System Constraints

- Development language shall be Java.
- Database shall be MySQL.
- The implementation shall demonstrate object-oriented programming concepts.
- The application shall include a graphical user interface.
- The Java application shall communicate with MySQL for persistent storage.

24. Project-Specific Academic Rules

24.1 Attendance

The required project dataset shall support 15 theory sessions and 15 practical sessions per subject.

24.2 Medical

Medical submissions shall be recorded with approval status.

24.3 CA Eligibility

CA $\geq$ 40% is required for eligibility for the final examination.

24.4 Marks

All marks shall be recorded out of 100.

24.5 Grades and GPA

Grades shall follow the required grading scheme. The system shall calculate grade points, SGPA and CGPA and support individual and batch-level result views.

25. Requirement Traceability Matrix

Requirement Area	Admin	Lecturer	Technical Officer	Undergraduate
Login	✓	✓	✓	✓
Profile Management	✓	✓	✓	✓
User Management	✓	—	—	—
Course Management	✓	View	View	View
Course Materials	—	✓	—	View
Marks	—	✓	—	View
Grades	—	✓	—	View
GPA	—	✓	—	View
Eligibility	—	✓	—	View/if permitted
Attendance	View	View	✓	View
Medical	View	View	✓	View
Notices	✓	View	View	View
Timetable	✓	View	View	View
Logout	✓	✓	✓	✓

26. Conclusion

The Faculty of Technology Academic Management System is a centralized, role-based platform for managing academic information within the Faculty of Technology. It supports Admin, Lecturer, Technical Officer and Undergraduate roles.

The system covers user profiles, courses, course materials, attendance, medical records, marks, grades, eligibility, GPA, notices and timetables. The requirements in this SRS provide the foundation for UML diagrams, ER/database design, Java implementation, GUI development, testing and deployment.

As an Object-Oriented Programming Practicum project, the implementation should demonstrate classes and objects, inheritance, abstraction, polymorphism, encapsulation, exception handling, database handling and GUI development.